Question 1

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```
data = read.csv("Quine.csv", stringsAsFactors = TRUE)
head(data)
```

```
##    Ethnicity Sex Grade Learner Days.Absent
## 1 Aboriginal   M    G6 Average           2
## 2 Aboriginal   M    G6 Average          11
## 3 Aboriginal   M    G6 Average          14
## 4 Aboriginal   M    G6  Slower           5
## 5 Aboriginal   M    G6  Slower           5
## 6 Aboriginal   M    G6  Slower          13
```

a)

```
model.a = glm(Days.Absent~.^2, data=data, family="poisson")
summ.a = summary(model.a)
summ.a
```

```
##
## Call:
## glm(formula = Days.Absent ~ .^2, family = "poisson", data = data)
##
## Coefficients: (1 not defined because of singularities)
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                           3.14854    0.14069  22.380  < 2e-16 ***
## EthnicityNon-Aboriginal               0.09016    0.13095   0.689   0.4911
## SexM                                 -0.67309    0.14823  -4.541 5.60e-06 ***
## GradeG7                              -0.17289    0.15004  -1.152   0.2492
## GradeG8                               0.28708    0.14570   1.970   0.0488 *
## GradeG9                              -0.15290    0.11898  -1.285   0.1987
## LearnerSlower                        -0.21608    0.14558  -1.484   0.1377
## EthnicityNon-Aboriginal:SexM          0.43902    0.09208   4.768 1.86e-06 ***
## EthnicityNon-Aboriginal:GradeG7      -0.92889    0.14657  -6.337 2.34e-10 ***
## EthnicityNon-Aboriginal:GradeG8      -1.33398    0.13504  -9.879  < 2e-16 ***
## EthnicityNon-Aboriginal:GradeG9      -0.11242    0.13478  -0.834   0.4042
## EthnicityNon-Aboriginal:LearnerSlower -0.26415    0.11378  -2.322   0.0203 *
## SexM:GradeG7                         -0.05565    0.16303  -0.341   0.7328
## SexM:GradeG8                          1.09942    0.15281   7.195 6.26e-13 ***
## SexM:GradeG9                          1.15949    0.13859   8.366  < 2e-16 ***
```

```
## SexM:LearnerSlower                    -0.04143    0.13718  -0.302   0.7627
## GradeG7:LearnerSlower                  0.13019    0.15688   0.830   0.4066
## GradeG8:LearnerSlower                 -0.37340    0.14563  -2.564   0.0103 *
## GradeG9:LearnerSlower                       NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2073.5  on 145  degrees of freedom
## Residual deviance: 1368.7  on 128  degrees of freedom
## AIC: 1993.1
##
## Number of Fisher Scoring iterations: 5
```

First, one coefficient (GradeG9:LearnerSlower) cannot be estimated, because there are no Average learners in Grade 9. Second, and most critically, the model exhibits severe overdispersion, the residual deviance of 1368.7 on 128 degrees of freedom gives phihat = 10.7, far exceeding 1.

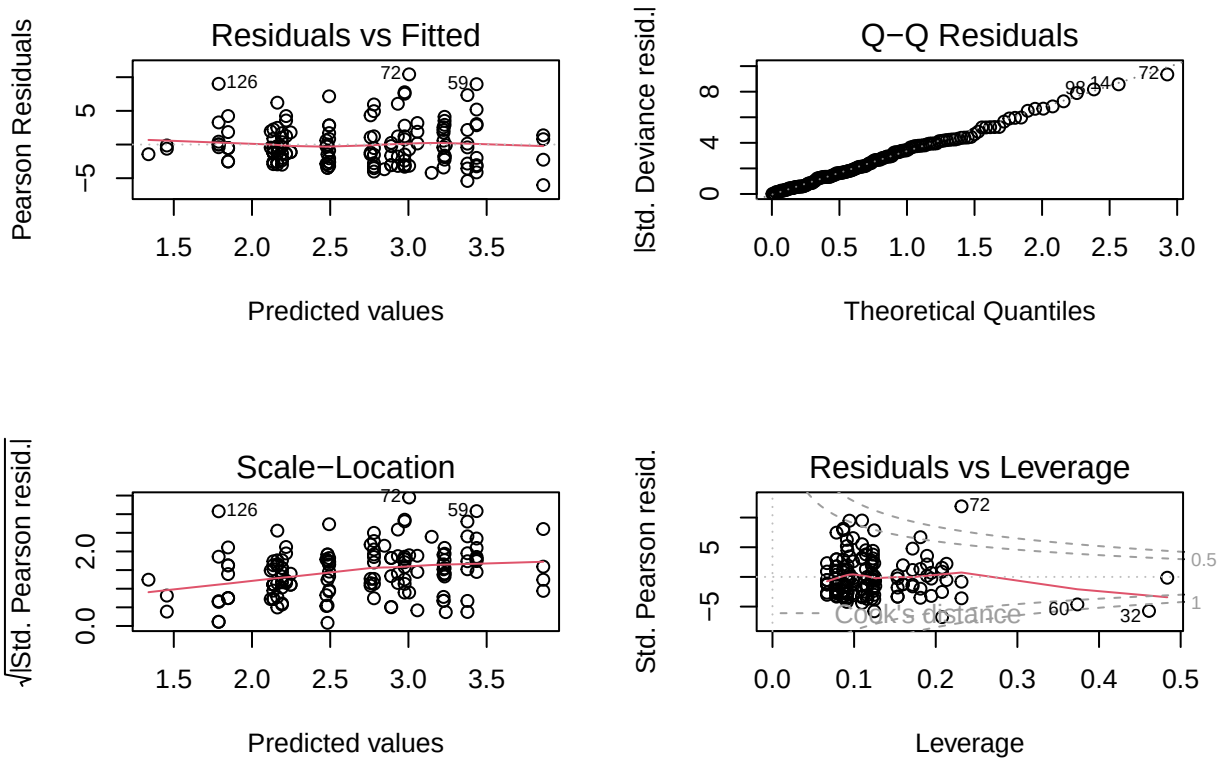
```
pvalue = pchisq(model.a$deviance, df=model.a$df.residual, lower.tail=FALSE)
pvalue
```

```
## [1] 1.458602e-206
```

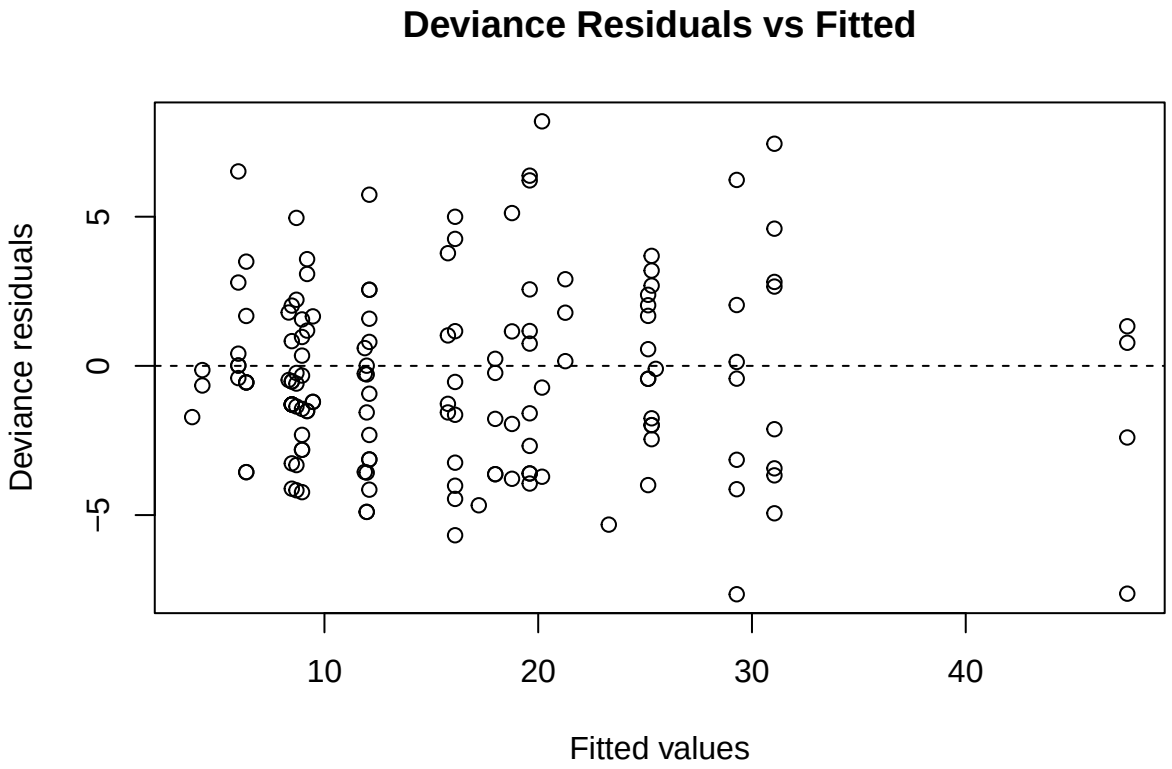
Conclusion: The p-value of 1.458602e-206 is way below 0.05, therefore we reject the null hypothesis H0 that the model with two-way interactions fits the data adequately. There is significant evidence that the model Days.Absent ~ .:Days.Absent is not a good fit.

b)

```
par(mfrow = c(2, 2))
plot(model.a)
```



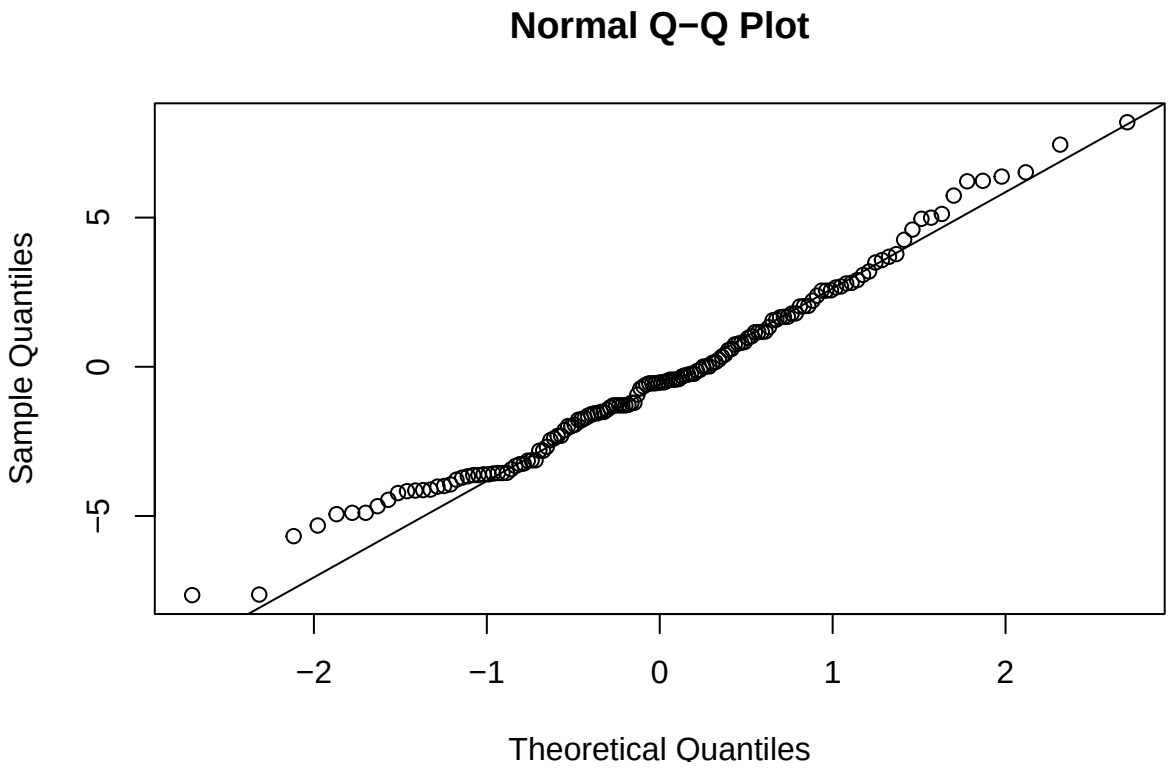

```
# Deviance residuals vs Fitted
plot(model.a$fitted.values, residuals(model.a, type = "deviance"),
     xlab = "Fitted values", ylab = "Deviance residuals",
     main = "Deviance Residuals vs Fitted")
abline(h = 0, lty = 2)
```



The

residuals show no clear systematic trend around zero. However, the spread is not constant, variability is notably larger at smaller fitted values (around 5–15) and there are several extreme residuals (beyond +-5) throughout the range.

```
# QQ plot of deviance residuals
qqnorm(residuals(model.a, type = "deviance"))
qqline(residuals(model.a, type = "deviance"))
```



The central bulk of points follows the line reasonably well. Both tails deviate substantially from the diagonal, the lower tail curves well below the line and the upper tail extends far above it.

Both plots consistently confirm the overdispersion already identified. The mean structure (covariates and interactions) appears reasonably specified, but the Poisson variance assumption is clearly violated.

c)

```
drop1(model.a, test="Chisq")
```

```
## Single term deletions
##
## Model:
## Days.Absent ~ (Ethnicity + Sex + Grade + Learner)^2
##                    Df Deviance    AIC     LRT  Pr(>Chi)
## <none>                  1368.7 1993.2
## Ethnicity:Sex       1   1391.6 2014.1  22.916 1.692e-06 ***
## Ethnicity:Grade     3   1497.4 2115.9 128.773 < 2.2e-16 ***
## Ethnicity:Learner   1   1374.1 1996.6   5.459   0.01947 *
```

```
## Sex:Grade          3   1518.2 2136.7 149.568 < 2.2e-16 ***
## Sex:Learner        1   1368.8 1991.2   0.091   0.76276
## Grade:Learner      2   1380.3 2000.8  11.611   0.00301 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion The only interaction that appears removable is Sex:Learner (LRT = 0.09, p = 0.763). All other interactions are significant and should be retained. However, given the overdispersion, these conclusions may not be reliable.

d)

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
mv <- data %>%
  group_by(Ethnicity, Sex, Grade, Learner) %>%
  summarise(
    mean_absent = mean(Days.Absent),
    var_absent = var(Days.Absent),
    n = n(),
    .groups = "drop"
  )
mv
```

```
## # A tibble: 28 x 7
##    Ethnicity  Sex   Grade Learner mean_absent var_absent     n
##    <fct>      <fct> <fct> <fct>         <dbl>      <dbl> <int>
##  1 Aboriginal F     G6    Average         3         NA       1
##  2 Aboriginal F     G6    Slower         21.2      314.      4
##  3 Aboriginal F     G7    Average        22.6      349.     10
##  4 Aboriginal F     G7    Slower         11.4       42.8     5
##  5 Aboriginal F     G8    Average        36.4      703.      8
##  6 Aboriginal F     G8    Slower          2         NA       1
##  7 Aboriginal F     G9    Slower         14.6      221.      9
##  8 Aboriginal M     G6    Average         9         39       3
##  9 Aboriginal M     G6    Slower         13         64.5     5
## 10 Aboriginal M     G7    Average         9         27       3
## # i 18 more rows
```

```r
mv <- aggregate(Days.Absent ~ Ethnicity + Sex + Grade + Learner,
                data = data,
                FUN = function(x) c(mean = mean(x), var = var(x)))

mv <- do.call(data.frame, mv)
mv
```

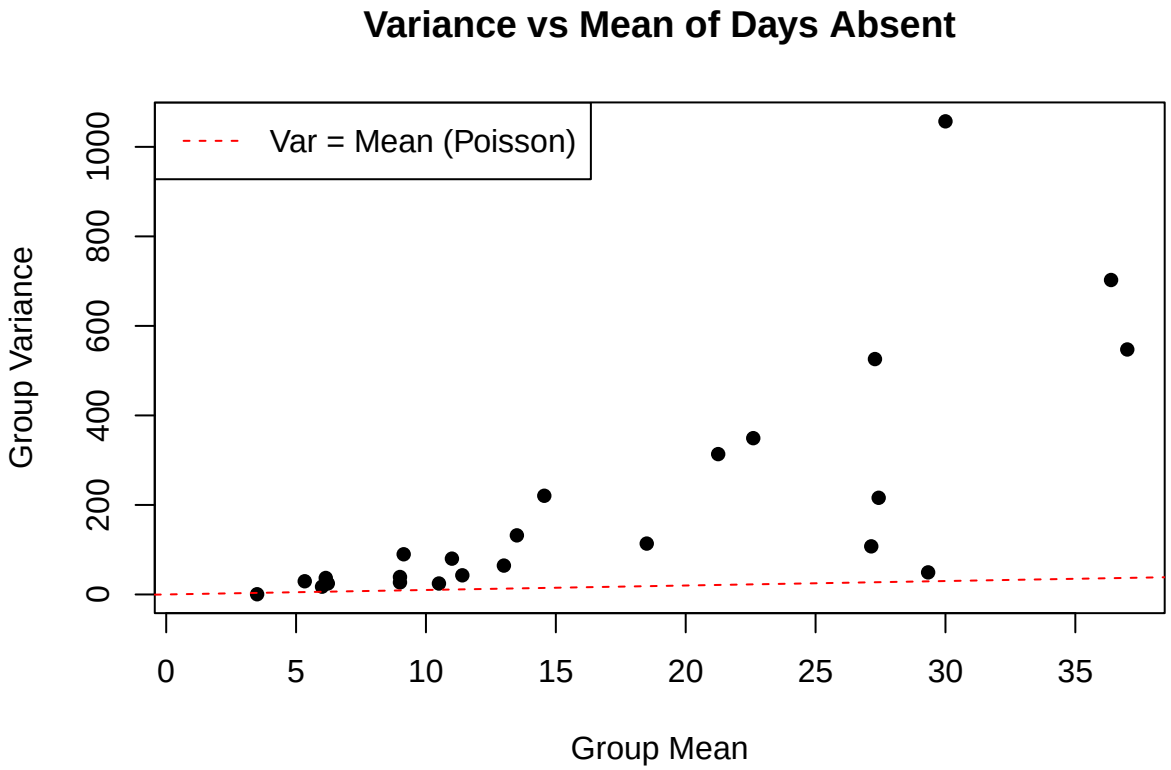
```
##          Ethnicity Sex Grade Learner Days.Absent.mean Days.Absent.var
## 1       Aboriginal   F    G6 Average         3.000000              NA
## 2  Non-Aboriginal   F    G6 Average        25.000000              NA
## 3       Aboriginal   M    G6 Average         9.000000        39.00000
## 4  Non-Aboriginal   M    G6 Average        30.000000      1057.00000
## 5       Aboriginal   F    G7 Average        22.600000       349.15556
## 6  Non-Aboriginal   F    G7 Average         6.000000        17.40000
## 7       Aboriginal   M    G7 Average         9.000000        27.00000
## 8  Non-Aboriginal   M    G7 Average         6.142857        36.80952
## 9       Aboriginal   F    G8 Average        36.375000       702.55357
## 10 Non-Aboriginal   F    G8 Average         6.222222        24.69444
## 11      Aboriginal   M    G8 Average        37.000000       547.33333
## 12 Non-Aboriginal   M    G8 Average        29.333333        49.33333
## 13      Aboriginal   F    G6  Slower        21.250000       313.58333
## 14 Non-Aboriginal   F    G6  Slower        18.500000       113.66667
## 15      Aboriginal   M    G6  Slower        13.000000        64.50000
## 16 Non-Aboriginal   M    G6  Slower         5.333333        29.46667
## 17      Aboriginal   F    G7  Slower        11.400000        42.80000
## 18 Non-Aboriginal   F    G7  Slower        11.000000        80.00000
## 19      Aboriginal   M    G7  Slower        10.500000        24.50000
## 20 Non-Aboriginal   M    G7  Slower         3.500000         0.50000
## 21      Aboriginal   F    G8  Slower         2.000000              NA
## 22 Non-Aboriginal   F    G8  Slower         1.000000              NA
## 23      Aboriginal   M    G8  Slower        27.428571       215.95238
## 24 Non-Aboriginal   M    G8  Slower         9.142857        89.80952
## 25      Aboriginal   F    G9  Slower        14.555556       220.52778
## 26 Non-Aboriginal   F    G9  Slower        13.500000       132.05556
## 27      Aboriginal   M    G9  Slower        27.142857       107.47619
## 28 Non-Aboriginal   M    G9  Slower        27.285714       525.90476
```

```r
names(mv)[5:6] <- c("mean_absent", "var_absent")

# Plot
plot(mv$mean_absent, mv$var_absent,
     xlab = "Group Mean",
     ylab = "Group Variance",
     main = "Variance vs Mean of Days Absent",
     pch  = 16)

# Poisson reference line: Var = Mean
abline(a = 0, b = 1, lty = 2, col = "red")

legend("topleft", legend = "Var = Mean (Poisson)",
       lty = 2, col = "red")
```



The plot provides direct empirical evidence that the Poisson assumption Var(Yi) = mui is violated across all covariate combinations.

e)

```
model.e = glm(Days.Absent~.^2, data=data, family="quasipoisson")
summ.e = summary(model.e)
summ.e
```

```
##
## Call:
## glm(formula = Days.Absent ~ .^2, family = "quasipoisson", data = data)
##
## Coefficients: (1 not defined because of singularities)
##                                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)                            3.14854    0.46069   6.834 3.01e-10 ***
## EthnicityNon-Aboriginal                0.09016    0.42881   0.210  0.83380
## SexM                                  -0.67309    0.48539  -1.387  0.16794
## GradeG7                               -0.17289    0.49133  -0.352  0.72551
## GradeG8                                0.28708    0.47712   0.602  0.54844
## GradeG9                               -0.15290    0.38960  -0.392  0.69538
## LearnerSlower                         -0.21608    0.47672  -0.453  0.65112
## EthnicityNon-Aboriginal:SexM           0.43902    0.30152   1.456  0.14783
## EthnicityNon-Aboriginal:GradeG7       -0.92889    0.47997  -1.935  0.05516 .
## EthnicityNon-Aboriginal:GradeG8       -1.33398    0.44220  -3.017  0.00308 **
## EthnicityNon-Aboriginal:GradeG9       -0.11242    0.44137  -0.255  0.79935
## EthnicityNon-Aboriginal:LearnerSlower -0.26415    0.37260  -0.709  0.47965
```

```
## SexM:GradeG7                          -0.05565    0.53386  -0.104  0.91714
## SexM:GradeG8                           1.09942    0.50040   2.197  0.02981 *
## SexM:GradeG9                           1.15949    0.45383   2.555  0.01179 *
## SexM:LearnerSlower                    -0.04143    0.44920  -0.092  0.92666
## GradeG7:LearnerSlower                  0.13019    0.51372   0.253  0.80035
## GradeG8:LearnerSlower                 -0.37340    0.47688  -0.783  0.43507
## GradeG9:LearnerSlower                       NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 10.72308)
##
##     Null deviance: 2073.5  on 145  degrees of freedom
## Residual deviance: 1368.7  on 128  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

The quasi-Poisson model is roughly identical in point estimates and deviance, but corrects for overdispersion by estimating phihat = 10.72

```
drop1(model.e, test="Chisq")
```

```
## Single term deletions
##
## Model:
## Days.Absent ~ (Ethnicity + Sex + Grade + Learner)^2
##                    Df Deviance scaled dev. Pr(>Chi)
## <none>                  1368.7
## Ethnicity:Sex       1   1391.6      2.1371 0.143774
## Ethnicity:Grade     3   1497.4     12.0090 0.007352 **
## Ethnicity:Learner   1   1374.1      0.5091 0.475529
## Sex:Grade           3   1518.2     13.9482 0.002977 **
## Sex:Learner         1   1368.8      0.0085 0.926553
## Grade:Learner       2   1380.3      1.0828 0.581920
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion

After using quasi_Poisson to fix the overdispersion issue, only Ethnicity:Grade and Sex:Grade are retained as significant interactions. The terms Ethnicity:Sex, Ethnicity:Learner, Grade:Learner, and Sex:Learner can all be considered for removal. This shows that the earlier Poisson analysis led to retaining interactions that are not genuinely supported by the data. The quasi-Poisson drop1 provides a much more honest basis for model simplification.

f)

```r
model.p.f = glm(Days.Absent ~ Ethnicity + Sex + Grade + Learner +
                      Ethnicity:Sex + Ethnicity:Grade + Ethnicity:Learner +
                      Sex:Grade + Grade:Learner,
                    data = data, family = "poisson")
summ.p.f = summary(model.p.f)
summ.p.f
```

```
##
## Call:
## glm(formula = Days.Absent ~ Ethnicity + Sex + Grade + Learner +
##     Ethnicity:Sex + Ethnicity:Grade + Ethnicity:Learner + Sex:Grade +
##     Grade:Learner, family = "poisson", data = data)
##
## Coefficients: (1 not defined because of singularities)
##                                      Estimate Std. Error z value Pr(>|z|)
## (Intercept)                           3.16615    0.12746  24.841  < 2e-16 ***
## EthnicityNon-Aboriginal               0.09070    0.13084   0.693  0.48819
## SexM                                 -0.70150    0.11433  -6.136 8.46e-10 ***
## GradeG7                              -0.18890    0.13982  -1.351  0.17669
## GradeG8                               0.27170    0.13605   1.997  0.04581 *
## GradeG9                              -0.14715    0.11754  -1.252  0.21062
## LearnerSlower                        -0.23890    0.12412  -1.925  0.05426 .
## EthnicityNon-Aboriginal:SexM          0.44103    0.09189   4.799 1.59e-06 ***
## EthnicityNon-Aboriginal:GradeG7      -0.92767    0.14649  -6.333 2.41e-10 ***
## EthnicityNon-Aboriginal:GradeG8      -1.33504    0.13502  -9.888  < 2e-16 ***
## EthnicityNon-Aboriginal:GradeG9      -0.11102    0.13471  -0.824  0.40985
## EthnicityNon-Aboriginal:LearnerSlower -0.26728    0.11338  -2.357  0.01840 *
## SexM:GradeG7                         -0.03978    0.15401  -0.258  0.79618
## SexM:GradeG8                          1.12251    0.13235   8.482  < 2e-16 ***
## SexM:GradeG9                          1.14549    0.13051   8.777  < 2e-16 ***
## GradeG7:LearnerSlower                 0.14681    0.14659   1.002  0.31657
## GradeG8:LearnerSlower                -0.38559    0.14003  -2.754  0.00589 **
## GradeG9:LearnerSlower                      NA         NA      NA       NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 2073.5  on 145  degrees of freedom
## Residual deviance: 1368.8  on 129  degrees of freedom
## AIC: 1991.2
##
## Number of Fisher Scoring iterations: 5
```

```r
anova(model.p.f, model.a, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model 1: Days.Absent ~ Ethnicity + Sex + Grade + Learner + Ethnicity:Sex +
##     Ethnicity:Grade + Ethnicity:Learner + Sex:Grade + Grade:Learner
## Model 2: Days.Absent ~ (Ethnicity + Sex + Grade + Learner)^2
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

```
## 1       129     1368.8
## 2       128     1368.7  1  0.09112   0.7628
```

Conclusion

Dropping Sex:Learner causes a negligible increase in deviance. The p-value of 0.763 confirms that this term contributes nothing to the model. The reduced Poisson is statistically equivalent to the full model while being more having less covariates.

```
model.qp.f <- glm(Days.Absent ~ Ethnicity + Sex + Grade + Learner +
                      Ethnicity:Grade + Sex:Grade,
                    data = data, family = "quasipoisson")
summ.qp.f = summary(model.qp.f)
summ.qp.f
```

```
##
## Call:
## glm(formula = Days.Absent ~ Ethnicity + Sex + Grade + Learner +
##     Ethnicity:Grade + Sex:Grade, family = "quasipoisson", data = data)
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      3.19550    0.32606   9.800  < 2e-16 ***
## EthnicityNon-Aboriginal          0.15313    0.33133   0.462  0.64472
## SexM                            -0.47567    0.33190  -1.433  0.15415
## GradeG7                         -0.14579    0.36491  -0.400  0.69014
## GradeG8                          0.18634    0.37402   0.498  0.61916
## GradeG9                         -0.08614    0.38979  -0.221  0.82544
## LearnerSlower                   -0.45575    0.18145  -2.512  0.01321 *
## EthnicityNon-Aboriginal:GradeG7 -0.97778    0.45115  -2.167  0.03199 *
## EthnicityNon-Aboriginal:GradeG8 -1.23650    0.42332  -2.921  0.00410 **
## EthnicityNon-Aboriginal:GradeG9 -0.18097    0.42083  -0.430  0.66786
## SexM:GradeG7                    -0.05498    0.49883  -0.110  0.91240
## SexM:GradeG8                     0.89849    0.42000   2.139  0.03424 *
## SexM:GradeG9                     1.13962    0.42388   2.689  0.00809 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 10.88183)
##
##     Null deviance: 2073.5  on 145  degrees of freedom
## Residual deviance: 1410.1  on 133  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```

```
anova(model.qp.f, model.e, test = "F")
```

```
## Analysis of Deviance Table
##
## Model 1: Days.Absent ~ Ethnicity + Sex + Grade + Learner + Ethnicity:Grade +
##     Sex:Grade
```

```
## Model 2: Days.Absent ~ (Ethnicity + Sex + Grade + Learner)^2
##   Resid. Df Resid. Dev Df Deviance      F Pr(>F)
## 1       133     1410.0
## 2       128     1368.7  5   41.386 0.7719 0.5717
```

Conclusion

Despite dropping 5 interaction terms (Ethnicity:Sex, Ethnicity:Learner, Sex:Learner, Grade:Learner, and one aliased term), the F-test is non-significant ($p = 0.572$). This means the dropped interactions are jointly non-significant, the reduced quasi-Poisson fits just as well as the full model.

Both reduced models are statistically indistinguishable from their full counterparts. The reduced quasi-Poisson is the most defensible final model, it is the most parsimonious, corrects for overdispersion, and retains only the two interactions (Ethnicity:Grade and Sex:Grade) that are genuinely supported by the data.